

Fluoride-Releasing Materials

Deniz Cakir-Ustun
Nathaniel C. Lawson
John O. Burgess

Patient assessment for oral problems should always begin with a caries risk assessment, which differentiates patients into categories of high, moderate, and low risk (see chapter 5) (Fig 14-1). Treatment is based on this assessment and is specialized by category. High-risk patients need specialized treatment prior, during, and after definitive restorative care. Fluoride is an important adjunct in caries prevention. In addition to professionally applied fluorides and fluoride-containing dentifrices, fluorides may be introduced into the oral environment with fluoride-releasing restorative materials. Fluoride plays several significant roles in any caries-prevention program. These include the formation of fluorapatite, which is more acid resistant than hydroxyapatite, the enhancement of remineralization, interference with ionic bonding during pellicle and plaque formation, and the inhibition of microbial growth and metabolism.[1] High concentrations of fluoride (such as those used in tray applications of fluoride at 12,300 ppm)[2] are bactericidal, while lower concentrations enhance remineralization; all levels produce fluorapatite crystals. Even when fluoride is present at concentrations as low as 1 µmol/L (approximately 0.02 ppm F), the oral fluids (saliva, plaque fluid) are supersaturated

with fluorapatite, inducing the precipitation of the mineral phase fluorapatite in the tooth structure.[3] Therefore, fluoride-releasing materials are essential in treating the high-caries-risk patient.



Fig 14-1 A patient with high caries risk.

Caries is a multifactorial disease with a bacterial etiology.[4] Because the bacteria producing this disease are introduced into the oral cavity by transfer from an infected host, a possible method for preventing the disease may be by blocking that transfer.[5] However, once established in the oral biofilm, caries-causing bacteria are not easily removed. Dental caries results from bacteria that metabolize sucrose or other cariogenic sugars (fructose and glucose) and secrete organic acids (lactic, propionic, and formic) that cause the loss of mineral ions (calcium and phosphates) from the tooth (*demineralization*).[6] Minerals lost by this method can be replaced during periods of neutral pH (*remineralization*) from calcium and phosphates in the saliva.

Remineralization is a repair process that relies on calcium and phosphate ions to rebuild a new surface on the remnants of crystals present after demineralization. Remineralization is facilitated by fluoride and can arrest carious demineralization in enamel by the formation of a hard and less permeable outer surface. Fluorides are most effective in inducing remineralization on smooth surfaces of teeth.[7] Calcium levels may be the dominant factor in the remineralization process, as the ratio of components required for remineralization is 10 calcium ions to 6 phosphate ions to 2 fluoride or hydroxyl ions. Periods of demineralization and remineralization make up a continuous cycle by which minerals in tooth structure are removed and replaced. If the balance is tipped toward demineralization, caries lesions develop.

The caries process in dentin is similar to that in enamel, except that dentin demineralization begins at a higher pH (6.4, compared with 5.5 for enamel) and

proceeds about twice as rapidly because dentin has only half the mineral content of enamel. Low fluoride levels are insufficient to initiate dentin remineralization but are adequate to facilitate enamel remineralization.[8] Fluoride ion concentration in saliva is low, averaging about 0.03 ppm (1.6 μmol/L) in the normal subject.[9] The balance of mineral uptake and loss in enamel is shifted from net demineralization to net remineralization if adequate calcium is present.[10] Restorative materials that release fluoride are often recommended for caries on root surfaces, because root structure is primarily composed of dentin and these lesions require significantly greater amounts of fluoride than enamel caries lesions to promote remineralization.

Individuals with high-caries-risk profiles are those with frequent carbohydrate intake, reduced salivary flow, increased plaque retention, low fluoride exposure during tooth formation, and high bacteria counts. Fluoride from fluoridated drinking water is filtered through the body and appears in the saliva. A normal unstimulated salivary flow rate is approximately 0.3 mL/min, while stimulated salivary flow ranges from 1.5 to 2.5 mL/min.[9] Low salivary flow is often associated with *(1)* medications such as antihypertensives, antidepressants, and anticholinergics, *(2)* a history of head and neck irradiation, or *(3)* Sjögren syndrome.[11,12] Low salivary flow reduces the bicarbonate, calcium, and phosphate ions that are typically provided by the saliva, thus reducing its buffering ability and its ability to induce remineralization of demineralized tooth structure.

Exposure to high levels of fluoride during tooth formation produces fluorapatite crystals in tooth structure. Fluorapatite crystals have a critical pH of 4.5, allowing fluorapatite to resist acid demineralization significantly better than hydroxyapatite, which has a critical pH of 5.5.[13]

It is clear that fluoride plays a significant role in caries prevention. Unfortunately, patient compliance in caries-control programs requiring home fluoride applications is often poor. As an alternative, fluoride-releasing materials can supply levels of fluoride that may provide a measure of caries prevention, without relying on patient compliance. Therefore, fluoride-releasing materials can be an essential element in treating the high-caries-risk patient. This chapter discusses fluoride-releasing materials, their effectiveness in inhibiting recurrent caries, and their clinical longevity.

The Fluoride-Releasing Materials Continuum

The first popular fluoride-releasing tooth-colored restorative material was silicate

cement. This material had poor adhesion to the tooth, high intraoral solubility, poor mechanical properties, and did not survive well in the oral environment. However, recurrent caries lesions associated with these restorations were rare. This anticaries effect was associated with the fluoride released from the silicate cement.[14]

Fluoride-releasing materials may be classified into five categories[15] based on similarities in physical, mechanical, and setting properties. These include *(1)* resin composite, *(2)* compomer, *(3)* giomer, *(4)* resin-modified glass ionomer (RMGI), and *(5)* conventional glass ionomer (Fig 14-2). Fluoride-releasing resin composites are on one end of the continuum and release the least fluoride, while conventional glass ionomers are on the other end and demonstrate the highest levels of fluoride release. Compomers are more similar to resin composites, and RMGIs are more similar to conventional glass ionomers. Giomers are a new class of material and are positioned between resin-modified glass ionomers and compomers.

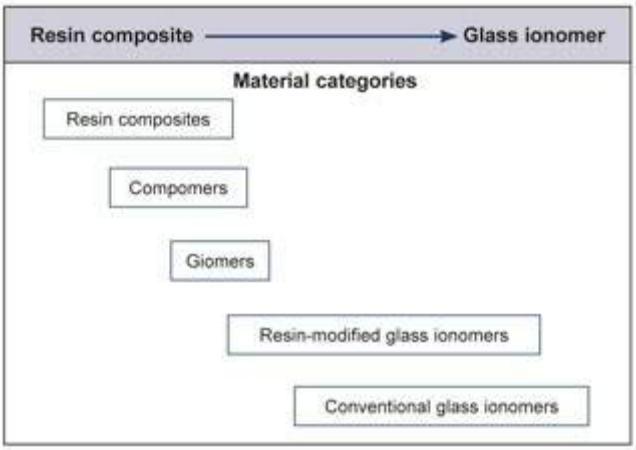


Fig 14-2 The fluoride-releasing material continuum. Fluoride release and recharge increase from left to right among the materials. Locations on the continuum are characterized by compositional variants, curing mechanisms, and mechanical and physical properties.

Table 14-1 compares the mechanical and physical properties of the different categories of fluoride-releasing materials. Table 14-2 describes the water content, supplied form, and setting mechanism (acid-base reaction or free radical polymerization) for these materials. Materials that are powder-liquid must have the powder mixed with the liquid to initiate set; paste-paste materials must have the two pastes mixed to initiate set; visible light–cured (VLC) materials must be illuminated by an appropriate curing light. Box 14-1 lists representative products in each category of the continuum.

Table 14-1	**Mechanical properties of materials on the fluoride-releasing materials continuum[16–24]**

	Material class				
	Glass ionomer	RMGI	Giomer	Compomer	Resin composite
Flexural strength (MPa)	30–67	60–93	115	94–115	76–135
Compressive strength (MPa)	99–198	143–184	325	217–295	217–315
Diametral tensile strength (MPa)	12	17–21	52	33–52	23–58
Shear bond strength (MPa)	4–8	7–13	12	11–13	17–30
Fluoride release	High	High	Moderate	Moderate	Low
Fluoride recharge	High	High	High	Moderate	Low

Table 14-2 Composition, form supplied, and setting reaction of the materials in the fluoride-releasing materials continuum

	Material class				
	Glass ionomer	RMGI	Giomer	Compomer	Resin composite
Composition	Contains water	Contains water	No water	No water	No water
Form	Powder-liquid	Powder-liquid or paste-paste	VLC paste	VLC paste	VLC paste or paste-paste
Setting reaction	Acid-base only	Acid-base and free radical	Free radical only	Free radical only	Free radical only

Box 14-1 Representative products in each category of the fluoride-releasing material continuum

Glass ionomer	RMGI	Giomer	Compomer	Resin composite
Ketac Fil*	Photac Fil*	Beautifil II[II]	Dyract AP†	Heliomolar‡ (contains fluoride)
				Tetric‡(contains

Fuji II[§]	Fuji II LC[§]		Compoglass[‡]	fluoride)
Ketac Molar*	Vitremer*		F2000*	SureFil[†] (contains fluoride)
Fuji IX[§]	Ketac Nano*			Z100* (No fluoride)

*3M ESPE

[†]Dentsply/Caulk

[‡]Ivoclar Vivadent

[§]GC America

[II]Shofu

The longevity and success of dental restorations depend on their ability to bond as well as seal the marginal interface (*adhesion*). This seal is affected by several of the properties of the restorative material, namely dimensional change during curing and the mismatch in thermal expansion between the material and the tooth. Both work to compromise the bond and seal. Glass ionomers chemically bond to dentin by an ionic bond with hydroxyapatite. The bond of conventional composites, compomers, and giomers to dentin is through micromechanical interlocking with collagen fibrils (*hybrid layer formation*) mediated by a dentin adhesive (see chapter 9). RMGIs contain components of glass ionomers (fluoro-aluminosilicate glasses and polyacrylic acid) as well as resin composites (photo- or chemical initiators and methacrylate monomers).[25] Because of their hybrid nature, RMGIs bond to dentin through both an ionic bond between polyacrylic acid and hydroxyapatite and mechanical interlocking with collagen and the resin monomer.[26] At the same time that this bond is being formed to the tooth, the material is being pulled away from the tooth surface as a result of contraction that occurs during the setting stage (*polymerization/setting shrinkage*). This contraction produces stress on the newly forming interfacial bond, leading to a loss of adhesion to the cavity walls when the tensile shrinkage stresses exceed the bond strength. This loss of adhesion compromises marginal seal, thereby increasing staining and postoperative cold sensitivity and potentially resulting in marginal caries.[27]

Gerdolle et al[28] measured polymerization shrinkage of four materials (RMGI, compomer, resin composite, and Ormocer) in vitro. Shrinkage values of materials ranked as follows: RMGI > compomer > Ormocer > resin composite. Another study by Attin et al[29] evaluated initial curing shrinkage of six RMGIs, a hybrid resin composite, and an autocured glass ionomer. Their study also showed that most of the RMGIs tested exhibited greater shrinkage than the hybrid composite or the autocured

glass ionomer. However, glass ionomers generate less polymerization shrinkage stress than do resin composites.[30] This factor, combined with water sorption by glass ionomers, can help to lessen the effects of polymerization shrinkage and reduce the probability of marginal gap formation.[31] Even when the bond resists the contraction forces, stresses may still be imposed on the interfacial bond during temperature changes in the mouth.

The coefficient of thermal expansion (CTE) measures the dimensional change of a material per degree change in temperature. The difference in CTE between the tooth structure (8.3×10^{-6}/°C to 1¼ $\times 10^{-6}$/°C, dentin and enamel respectively) 32 and the restorative materials causes additional stress at the restoration margin, which may further contribute to marginal failure. The CTE of glass ionomers is similar to that of human enamel and dentin, whereas composites have CTEs approximately twice that of human enamel and dentin.[33]

Another important property of glass ionomers is fluoride release, which can potentially help to mitigate the effects of less-than-perfect marginal adaptation. However, fluoride release diminishes with time. Fortunately, glass ionomers have the ability to replenish leached fluoride when exposed to topically applied high fluoride–containing solutions. This property, known as *recharge*, has the potential to provide a continuous low concentration of fluoride in the saliva[34] because the replenished fluoride can be released back to the oral environment. An essential factor in quantifying the recharge ability of the glass ionomer is the amount of fluoride at the surface of the material, where it reaches its maximum concentration.[35]

Resin composites

Fluoride can be incorporated into resin composites through three mechanisms: *(1)* adding fluoride compounds such as sodium fluoride (NaF) or stannous fluoride (SnF_2) *(2)* incorporating fluoride-releasing fillers; or *(3)* integrating fluoride into the resin matrix.[36] Fluoride-releasing resin composites have superior mechanical properties, no inherent adhesive properties (they need a bonding agent to adhere to tooth structure), greater CTEs, and better wear resistance compared with other materials in the continuum. However, they also have the lowest fluoride release and are least capable of being recharged.

Glass ionomers

Glass ionomers are comprised of a polyalkenoic acid polymer matrix and ion-

leachable fluoroaluminosilicate glass filler particles. Carboxylic acid chains on the polyalkenoic acid form ionic bonds with calcium and aluminum ions leached from the glass filler (Fig 14-3) to cross-link the polymer and form a rigid structure. The carboxylic ions also chelate with calcium (Ca^{2+}) ions in hydroxyapatite, allowing the material to chemically bond to tooth structure.[37] Thus, these materials are adhesive, release comparatively high levels of fluoride, and have CTEs similar to that of tooth structure (Fig 14-4). Early brand-name glass ionomers are still widely used today despite their relatively poor mechanical properties and wear resistance. A newer generation of high-viscosity glass ionomers (Ketac Molar [3M ESPE] and Fuji IX [GC America]) has improved mechanical properties and provides higher levels of fluoride release compared with traditional glass ionomers.[15,39] Although wear resistance is improved, these materials are still inferior to resin composites and should not be used to restore load-bearing areas in the permanent dentition (Fig 14-5). Walls and Mather[39] reported a 1-year wear rate of 73 μm, which is significantly greater than the < 10 μm of wear per year reported for various resin composites.[41,42] Frankenberger et al_{43} reported 8% and 40% failure rates for Class 1 and Class 2 glass-ionomer restorations, respectively, after 24 months.

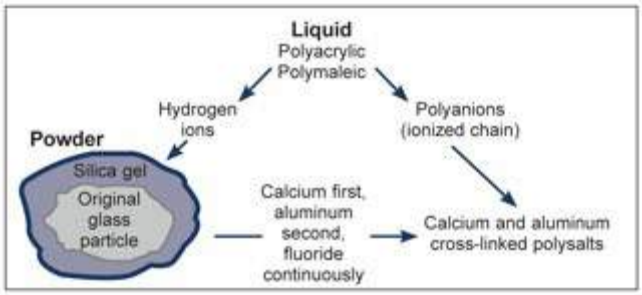


Fig 14-3 Schematic of the setting reaction of chemically cured glass ionomers.

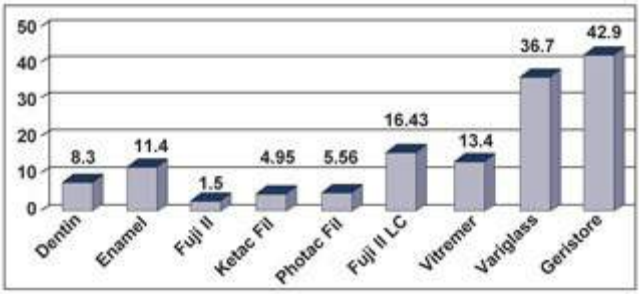


Fig 14-4 Thermal expansion (ppm/°C) of glass ionomers (Fuji II, Ketac Fil), RMGIs (Photac Fil, Fuji II LC, Vitremer), and compomers (Variglass [Dentsply/ Caulk], Geristore [DenMat]).[32,38]

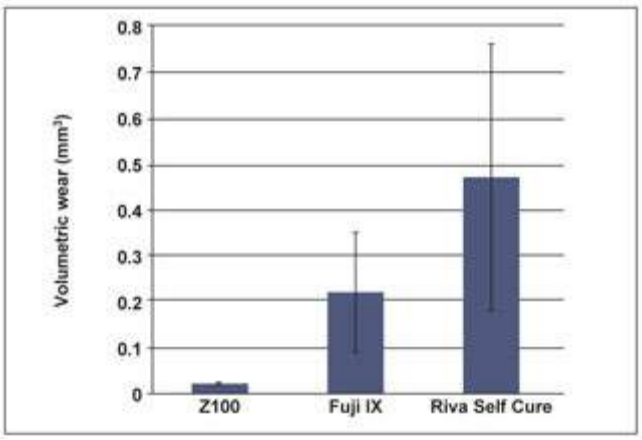


Fig 14-5 Volumetric wear of a composite (Z100) and two conventional glass ionomers (Fuji IX, Riva Self Cure [SDI]).[40]

Resin-modified glass ionomers

RMGIs contain elements of conventional glass ionomers and light-cured resins but have properties most similar to conventional glass ionomers. The matrix consists of polyalkenoic acids modified with unsaturated carbon bonds and/or mixed with (di)methacrylate monomer(s), allowing these materials to cure through acid-base–initiated ionic bonding and polymerization through free radical–initiated covalent bonding.[37] The amount of resin material in the different brands varies widely, because in recent years an effort has been made to increase the resin content to improve handling and strength of the RMGIs. RMGIs develop bond strength primarily through a chemical interaction with the hydroxyapatite in the tooth, but evidence for hybrid layer formation exists as well, especially for formulations with greater resin content.[44] RMGIs have been modified in several ways since the fluoride-releasing materials continuum was proposed. Improved manufacturing has led to smaller filler particles, resulting in a smoother restoration surface and increased fluoride release.[45] Recently, paste-paste RMGIs (those supplied as two pastes that are mixed together) have been marketed as luting cements and more recently as restorative materials. Paste-paste materials are easier to mix and place than those supplied in powder-liquid form. However, in one study, the paste-paste systems showed lower crown retention strength than powder-liquid systems.[46] RMGIs, like conventional glass ionomers, should not be used for restorations in occlusal load–bearing areas in the permanent dentition where heavy occlusal forces are present. However, the results of a 1-year clinical study suggest that RMGIs may be an acceptable short-term alternative to amalgam in pediatric restorations with respect to failure rate.[47]

Compomers

Compomers (polyacid-modified resin composites) are also blends of resin composites and conventional glass ionomers, created in an attempt to combine particular characteristics of both materials. However, they are primarily composite in composition, and their physical and mechanical properties are more similar to those of the fluoride-releasing resin composites. In this material, carboxylic groups are added to dimethacylate monomers in the matrix, and the fluoroaluminosilicate glass filler particles are partially silanized for bonding with the resin matrix.[37] Compomers require a dentin bonding system and acid etching of tooth structure to achieve a clinically acceptable bond.[48,49] These materials release more fluoride than resin composites but less than conventional glass ionomers or RMGIs. Their abrasion resistance is intermediate between RMGIs and resin composites. Compomer restorative materials and cements have undergone considerable improvements since their introduction. These improvements have produced materials with increased fluoride release and improved mechanical properties. Expansion due to water sorption was an early problem with compomers, but this problem is less severe in currently available products.[50] The monomer components of at least one of the available brands contains a small proportion of carboxylic acid functional groups, insufficient to confer water solubility on either the monomer or its polymer but potentially capable of promoting an acid-base neutralization reaction following sorption of water after polymerization, with the subsequent release of fluoride ions.[51]

Giomers

A new class of material has been introduced to the market that is termed *giomer* (glass-ionomer and polymer). Giomers use prereacted glass-ionomer (PRG) technology to form a stable phase of glass-ionomer fillers in a resin matrix. The PRG filler is made by reacting fluoride-containing glass with polyalkenoic acid in water. The PRG filler is then freeze-dried, milled to a desired size, silanized, and added to the resin matrix.[52] The advantage of giomers is the enhanced availability and accessibility of the fluoride within the PRG fillers compared with that in resin-based materials (such as compomers and certain resin composites). In a polymerized dimethacrylate resin matrix, which is mostly hydrophobic, fluoride transfer is dependent on water sorption and segmental mobility of highly cross-linked polymer chains. In giomer materials, the PRG fillers are more accessible to release fluoride and recharge with fluoride. Two types of PRG filler technology are available: *(1)*

surface-reacted PRG filler (S-PRG filler) and *(2)* fully reacted PRG filler (F-PRG filler). The difference between filler types is that only the ions on the surface of S-PRG filler particles leach from the glass and react with polyalkenoic acid. As a result, S-PRG fillers have only a thin outer layer of glass-ionomer phase surrounding an unreacted glass core, while the F-PRG fillers have been reacted more completely throughout.[52]

Giomer materials have fluoride recharge, biocompatibility, a smooth surface finish, excellent esthetics, and clinical durability, which have made them popular for restoration of root caries, noncarious cervical lesions, Class 5 cavities, and lesions in primary teeth.[51] A clinical trial of Class 1 and Class 2 giomer restorations in permanent teeth reported no failures of 41 restorations at the 8-year recall.[53] A recent in vitro study compared the fluoride release and recharge of two fluoride-containing resin composites, Gradia Direct X (GC America) and Tetric EvoCeram (Ivoclar Vivadent); a new giomer material, Beautifil II (Shofu); and a glass-ionomer cement, Fuji IX Extra (GC America), when aged in deionized water (pH 6.5) and lactic acid (pH 4.0) for 43 days. Fluoride release was measured during aging for 6 weeks. Each specimen was recharged in 5,000-ppm NaF solution for 5 minutes at 7 weeks of aging. Recharge was repeated weekly for 3 weeks and fluoride release measured 1, 3, and 7 days after each recharge. The cumulative fluoride released from the giomer into both media was substantially greater than the fluoride released from the other two composites, as was its recharge ability in both media. However, the glass-ionomer cement had the greatest fluoride release and recharge of all the materials.[54]

Fluoride Release

Several researchers have reported that fluoride released from restorative materials affects tooth structure. Decreased recurrent caries rates historically found around silicate restorations have been associated with their high level of fluoride release. Early in vivo work by Hals and Norderval[55] examined recurrent caries around Class 5 restorations and reported a lower occurrence of caries lesions around the silicate restorations. In 1957, Phillips and Swartz[56] measured the enamel solubility around fluoride-releasing restorative materials. Because the influence of water fluoridation in reducing enamel solubility was already well established, it was postulated that fluoride released from dental restorative materials could be incorporated into the tooth structure around the restorations, contributing to reduced enamel solubility.

This study demonstrated that fluoride present in silicate cements reduced the solubility of the adjacent enamel and suggested that fluoride in low concentrations could be added to dental materials to reduce enamel solubility.

In 1960, Norman et al[57] reported the fluoride content of powdered enamel exposed to fluoride released from porcelain, resin with added NaF, zinc phosphate cement, zinc phosphate cement with 10% calcium fluoride, 2% NaF, and silicate cements. Silicates released large amounts of fluoride for the first 24 hours, but the levels decreased with time. The enamel specimens exposed to the silicates had the greatest fluoride uptake (increase of 495%). The results of this study suggested that enamel exposed to materials releasing high levels of fluoride showed significant fluoride uptake. Furthermore, whenever large amounts of fluoride ions were present in solution, only a small percentage of the available fluoride ions (eg, 18% in the case of enamel exposed to silicate cement) was actually absorbed by the enamel. With smaller amounts of fluoride ions in the solution, a larger percentage was absorbed by the enamel. Even low levels of fluoride release for extended periods may be important for caries prevention, as it has been shown to be dependent on sustained low-level fluoride exposure rather than high-level, short-term bursts.[58]

Subsequently Norman et al[14] examined fluoride uptake by enamel slabs exposed to the same dental materials. The data from powdered enamel and intact enamel were generally in agreement. Most of the materials tested produced some measurable increase in enamel fluoride content, and fluoride increases correlated with reduced acid solubility of the intact enamel. Although the research confirmed that silicate cement released relatively high levels of fluoride, these materials ultimately tended to fail because of their high degree of solubility and poor mechanical properties.

The common denominator in this early research on the effects of various restorative materials on plaque and caries activity appeared to be the presence of fluoride.[59] Because fluoride was present in, and leached from, silicate cements, subsequent research was devoted to finding other materials with improved physical and mechanical properties that might release fluoride in a more predictable and clinically more successful manner. This preliminary work ultimately led to the development of glass-ionomer cement.

There are significant numbers of in vitro and in vivo studies that demonstrate the fluoride-releasing capability of glass ionomers.[50,60] Researchers have measured the fluoride release from four glass-ionomer cements in vitro and reported that the greatest release occurred on the first day; subsequently, it decreased sharply the

second day and gradually diminished over 3 weeks to a low-level, long-term release. After 1 year, all specimens were still releasing fluoride at daily levels of at least 0.5 ppm. Other studies have reported a "burst" of fluoride release, with high early release for 1 to 2 days followed by a rapid decline.[61–72] The early fluoride burst is most likely due to the availability at the material surface of fluoride ions from the matrix that were released during the reaction between the glass particles and polyalkenoate acid, as well as from soluble fluoride added to the glass-ionomer powder.[73] Prolonged fluoride release occurs as fluoride dissolves from the glass particles surrounded by an acidified hydrogel matrix[74] and then diffuses through the polymer matrix to be released from the surface in a time-dependent diffusion process.

In vivo research by Hatibovic-Kofman and Koch[75] and by Hattab et al[76] found significant increases of fluoride in saliva following the placement of glass-ionomer restorations. In the former study, salivary fluoride concentrations remained elevated 1 year after placement of glass-ionomer restorations (0.04 ppm before placement of restorations, 0.8 ppm 3 weeks after placement, and up to 0.3 ppm 1 year later).[75] In an in situ study, subjects wore maxillary appliances with four glass-ionomer (Ketac Fil, 3M ESPE) restorations every night.[76] The unstimulated salivary fluoride content was measured before insertion of the device and after overnight wear. In all subjects, salivary fluoride increased after wearing the appliance. Xu and Burgess[45] measured the compressive strength, fluoride-release, and fluoride-recharge profiles of 15 fluoride-releasing materials. The fluoride-release profiles of the glass-ionomer materials in this study are shown in Fig 14-6. The study demonstrated a negative (inverse) correlation between fluoride release and compressive strength (ie, materials that have a high level of fluoride release generally have lower strengths than materials with a low level of fluoride release).

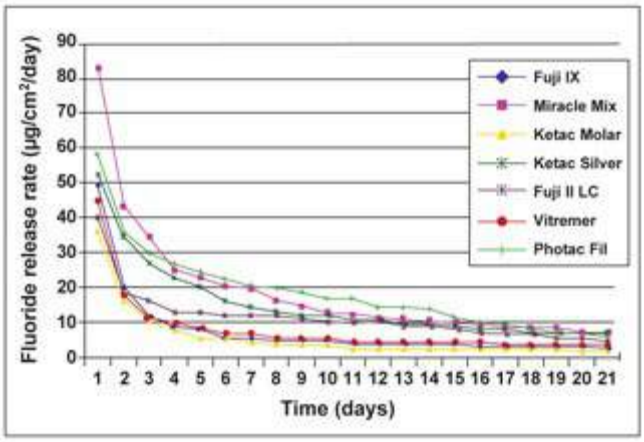


Fig 14-6 Fluoride-release profiles of some glass-ionomer restorative materials over 21 days45: conventional glass ionomers, Fuji IX and Ketac Molar; metal-reinforced glass ionomer, Miracle

Mix (GC America); silver-containing glass ionomer, Ketac Silver (3M ESPE); RMGIs, Fuji II LC, Vitremer (3M ESPE), and Photac Fil (3M ESPE).

Hsu et al[71] measured the fluoride released from an RMGI (Vitremer, 3M ESPE) and a high-viscosity conventional glass ionomer (Fuji IX) using a continuous flow rate apparatus adjusted to 20 mL/h. They reported that Vitremer released more fluoride than Fuji IX over an 8-hour period. In addition, the fluoride-release rate decreased rapidly after recharging, and most of the fluoride was released within 6 hours. This is a clinically significant finding because it indicates that the fluoride-releasing materials should be recharged with external neutral NaF daily to increase their fluoride release and reminer-alizing potential (Fig 14-7).

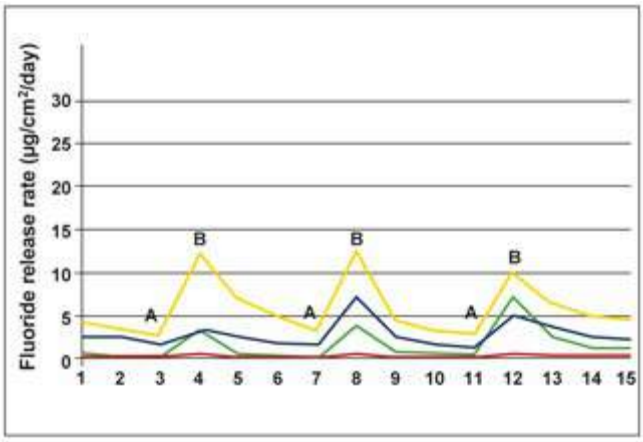


Fig 14-7 Fluoride-recharge profiles of fluoride-releasing restorative materials.[71] (A) Baseline amount of fluoride release prior to application of topical fluoride; (B) amount of fluoride release 1 day after recharge. Green—glass ionomer; yellow—RMGI; blue—compomer; red—composite.

Mousavinasab and Meyers[17] measured the amounts of fluoride released from four classes of fluoride-containing materials including three glass-ionomer cements (Fuji IX, Fuji VII, and Fuji IX Extra), one RMGI (Fuji II LC [GC America]), a compomer (Dyract Extra, Dentsply/Caulk) and a giomer (Beautifil). Cumulative fluoride release for up to 21 days was greatest from Fuji VII, followed by Fuji IX Extra, Fuji II LC, Fuji IX, Dyract Extra, and Beautifil, with the dimethacrylate resin–matrix materials—the compomer (Dyract) and giomer (Beautifil)—being significantly lower than the glass-ionomer materials. This work emphasizes the importance of the glass-ionomer matrix in determining the fluoride-releasing ability of glass-ionomer materials.

In a comprehensive study of fluoride-releasing materials, Cranfield et al[63] reported that pH influenced fluoride release in storage media, with a lower pH producing higher fluoride release, probably due to erosion of the glass-ionomer

surface. Moreau and Xu[77] also examined fluoride released from glass ionomers, a composite, and a compomer in vitro in varying pH solutions. They determined that these materials had increased fluoride release at a cariogenic pH—a useful property for anticaries effects.

Fluoride release from a restoration that can be incorporated into tooth structure and into the walls of the cavity preparation is perhaps the most important benefit of glass ionomer.[56,78,79] However, the fluoride release and incorporation into the tooth can be decreased and inhibited by a layer of adhesive if used beneath the RMGI to enhance bonding.[80,81] If applied to the surface of the restoration (ie, to reduce the potential for moisture loss during setting) the adhesive initially may reduce fluoride release from the surface and to whatever depth the adhesive penetrates into the surface porosity. However, the resin coating on restoration surfaces will in time be removed due to toothbrushing and mastication and therefore not have a significant impact on long-term fluoride release from the external surface of the restoration.[82,83]

Fluoride Recharging

Perhaps the most important variable in fluoride release is not the amount of fluoride released from the material initially after placement, because this declines rapidly with time (see Fig 14-6), but the ability of the material to be recharged with fluoride from external sources (see Fig 14-7). Forsten evaluated fluoride release[67] and uptake[84] by traditional glass ionomers to determine whether glass-ionomer materials could not only release fluoride but also take up fluoride from a fluoride-rich solution. He reported that after an initial high rate of fluoride release, a constant level occurred at about 3 weeks and that topical fluoride applications could recharge glass-ionomer restorations. Forsten[68,84] also reported a constant release rate of approximately 0.5 to 1.0 μg/mL during the second year for all glass ionomers except cermets (glass ionomers containing silver particles). Other investigators[79,85–87] have confirmed the recharge phenomenon when high fluoride–content solutions are applied to glass ionomers.

Compomers and RMGIs can be recharged as well.[88] RMGIs, such as Photac Fil (3M ESPE), Vitremer, and Fuji II LC, and conventional glass ionomers, such as Fuji IX and Ketac Molar demonstrate the greatest fluoride-recharge capacity. Fluoride-releasing resin composites, such as Heliomolar (Ivoclar Viva-dent) and Tetric Ceram (Ivoclar Vivadent), release little additional fluoride after being exposed to a fluoride-rich solution. Compomers, such as Dyract AP (Dentsyply/Caulk) and

Compoglass (Ivoclar Vivadent), have a recharge capacity between those of RMGIs and resin composites but are most similar to RMGIs. For materials with recharge capability, the fluoride release remains at increased levels for less than a day after being recharged. Xu and Burgess[45] showed that the RMGIs as a group, had the greatest recharge, followed by compomers. Traditional glass ionomers were intermediate, and fluoride-releasing resin composites had the least recharge (see Fig 14-7). This recharge capability may be due to the microporosities present in glass ionomers (Fig 14-8) and RMGIs. A recent in vitro study[18] compared fluoride release from a giomer and a compomer after fluoride-recharge regimes and concluded that giomers have significantly greater fluoride release and uptake following recharging than compomers.

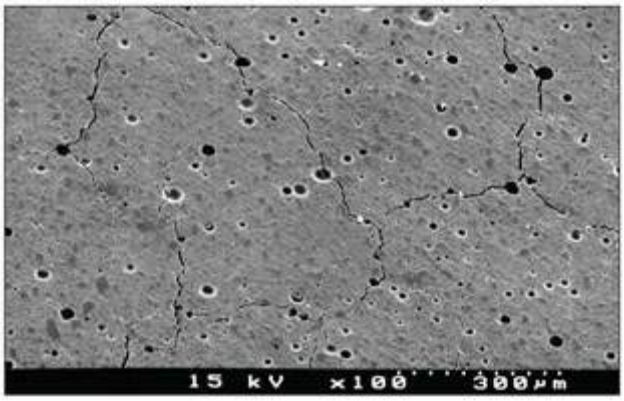


Fig 14-8 Photomicrograph of porosity in Fuji IX that allows fluoride-rich materials to be absorbed and released.

The ability of a material to act as a fluoride reservoir is dependent on the type and concentration of fluoridating agent, the frequency of recharge, and the solubility and type of restorative material.[18] Once the fluoride is depleted, it may be replenished from other fluoride sources, such as toothpastes, mouthrinses, or topical fluoride solutions. High-concentration fluoride solutions such as gels, when applied to fluoride-releasing materials, produce greater fluoride recharge than low fluoride–containing toothpastes. The increased fluoride release after recharge is short-lived, and recharge must be accomplished daily to maintain an elevated level. After recharging, fluoride ions are released from these materials by washout of fluoride from their surface or their pores.[89]

The pH of the topical fluoride used to recharge glass-ionomer restorations is important. Acidic topical fluoride solutions, such as acidulated phosphate fluoride solutions and other acidified fluoride preparations, degrade glass-ionomer materials and should be avoided.[90–92] RMGIs are more resistant to surface degradation than

conventional glass ionomers but still degrade when exposed to acids.[92] Resin composites are also degraded by frequent applications of acidic fluoride solutions, producing filler dislodgment and destruction of the filler–resin matrix interface.[93]

Antibacterial Effects of Fluoride

The high fluoride levels released from glass-ionomer restorative materials in the first few days after placement may be beneficial because of antibacterial effects. Initial microleakage often occurs after restoration placement.[94] Fluoride has three effects on bacteria: *(1)* inhibition of metabolism, *(2)* inhibition of growth, *(3)* and bacterial death. Fluoride is bactericidal in concentrations significantly greater than the amount released by fluoride-releasing materials. Thus, a direct bactericidal effect probably does not occur from fluoride released by restorative materials. However, fluoride levels produced by fluoride-releasing materials probably affect bacterial metabolism. By reducing bacterial acid production and altering the environment, fluoride release effectively changes the succession of bacteria necessary for acid production. This subtle mechanism is difficult to measure but may profoundly reduce acid production and tooth demineralization by cariogenic bacteria.

Fluoride released from dental materials is incorporated into bacteria[9] and inhibits bacterial acid production.[10] However, the fluoride in plaque on teeth adjacent to or several teeth distant from a glass-ionomer restoration has not been shown to be increased, emphasizing that the effect is very local. Protection provided by fluoride-releasing materials is probably confined to tooth structure immediately adjacent to the restoration, and one study demonstrated the effective zone to be about 1 mm from the restoration's margins.[85] Several studies have postulated that caries formed around a restoration may in fact be new lesions formed adjacent to previous restorations and not secondary caries formed from carious material left under the previous restoration.[95–98] Fluoride-releasing materials may play an important role in controlling the development of new lesions.[99]

Norman et al[59] examined the effects of fluoride-containing and other restorative materials on bacterial plaque. They compared plaque composition associated with amalgam, gold foil, cast gold (inlay), methyl methacrylate resin, and silicate cement restorations. They reported that comparable plaque developed on all tested materials with the exception of the silicate cement. Based on this research, it appeared that the fluoride component of the silicate cement somehow altered the

composition of the plaque, both at the margin of the restoration and on the tooth surface. Loyola-Rodriguez et al[100] observed similar inhibition of *Streptococcus mutans* by resin-modified and conventional glass-ionomer restorative materials. Forss et al[101,102] demonstrated a higher level of fluoride in plaque growing on glass ionomer than in plaque growing on resin composite restorations, with a significant reduction of *S mutans* near the glass-ionomer restorations.

In an in vitro study, Palenik et al[103] measured the inhibition by six glass-ionomer materials of microbial adherence and growth of five different bacteria typical in human plaque (*Actinomyces viscosus, Streptococcus mitis, Streptococcus mutans, Lactobacillus casei,* and *Streptococcus sanguis*). All glass ionomers inhibited growth and/or plaque adherence to some degree. The extent of inhibition varied according to the material used and the specific bacterial species. Growth inhibition was directly related to the amount of fluoride ion released; materials with a higher rate of release had a greater effect. McComb and Erickson[104] demonstrated that a pH of 2.6 was produced with freshly mixed glass ionomer, rising ultimately to 7.4 during the setting reaction. These investigators proposed that both the low pH and the fluoride release were responsible for the antibacterial effects. DeSchepper et al[105] used an agar diffusion assay method to determine the inhibitory effects on *S mutans* of 11 glass-ionomer cements and their powder and liquid components. They measured the zone of inhibition of bacterial growth and the fluoride levels in the surrounding agar. Because glass-ionomer liquids (polyacids) are acidic, they inhibited bacterial growth. The mixed materials released significant amounts of fluoride, and the inhibition zones correlated well with fluoride levels; high fluoride-releasing materials had greater zones of inhibition than low fluoride-releasing materials. However, in a subsequent study[106] measuring the antibacterial effects of light-cured glass ionomers, a neutral NaF control was included. Although Vitrebond (3M ESPE) and XR Ionomer (Kerr) produced significant inhibition zones, the neutral NaF produced no inhibition zone. When the pH of the highly acidic tested materials was adjusted upward to 5 by adding sodium hydroxide, none had antibacterial effects. The authors concluded that the mechanism of action for bacterial inhibition was probably a combination of fluoride release and low pH. A recent study measured bacterial growth in vitro on a glass ionomer, an RMGI, a compomer, a giomer, and a resin composite and showed that the glass ionomer, despite releasing the most fluoride, had the lowest antibacterial properties. The study suggests that either fluoride release is not the dominant mechanism for reducing bacterial growth and biofilm formation on a restorative material or that the fluoride concentration produced from fluoride-releasing materials is too low to be

effective.[107]

Bacteria in plaque located on glass-ionomer restorations are affected less as the age of the restoration increases. Bacterial composition of plaque on glass-ionomer restorations, up to a month after placement,[101] generally show a strong positive correlation between fluoride release and reduced *S mutans* counts. However, the data on older restorations, which release less fluoride and have a higher pH, are equivocal regarding bacterial counts. Svanberg et al[108] showed lower bacterial counts and reduced fluoride levels in the bacteria, while Van Dijken et al[109] reported decreased antibacterial effect of the restorative materials. The difference between findings in these studies may be due to differences in the materials and/or the plaque sampling. Svanberg used a silver-containing glass ionomer, and the silver release may have provided additional antibacterial action compared with that of the RMGI in the Van Dijken study. In the latter study, plaque samples were obtained from the surface of the restorations,[109] whereas plaque was obtained from the restoration margins in the Svanberg study.[108] The concentration of fluoride may be higher in the more protected environment of restoration margins. Also, Van Dijken discontinued oral hygiene procedures for 1 week to allow plaque to accumulate prior to sampling.[109] Eliminating the potential for fluoride recharge of the surface from fluoride-containing toothpaste during the week preceding sampling may also have contributed to the reduced antibacterial effects in the Van Dijken study.

Do Fluoride-Releasing Materials Inhibit Caries?

As previously stated, several mechanisms have been suggested for the anticaries effect of fluorides. These include the formation of fluorapatite, which is more acid resistant than hydroxyapatite; the enhancement of remineralization; the interference of ionic bonding during pellicle and plaque formation; and the inhibition of microbial growth and metabolism.[1] Fluoride released from restorative materials can inhibit development of caries lesions through all of these mechanisms, although it seems likely that enhancement of remineralization is the most important mechanism of action for fluoride released from restorative materials. Evidence for the caries-inhibiting effect of fluoride-releasing materials comes from studies of the incidence of caries lesions adjacent to orthodontic bands and brackets, direct filling materials in patients at high risk for caries, and areas around atraumatic restorative technique (ART) restorations.[110]

Although not directly applicable, the effects of orthodontic luting materials on

adjacent tooth structure may supply valuable information regarding fluoride-releasing restorative materials, because enamel demineralization frequently occurs adjacent to fixed orthodontic appliances. White spot lesions appear within a few weeks after appliance placement, particularly with ill-fitting bands.[111–113] An increased cariogenic challenge is introduced around orthodontic bands and brackets due to markedly higher plaque retention levels caused by food retention and hindered patient oral hygiene efforts. A rise in cariogenic bacteria (*S mutans* and *Lactobacillus*) has been documented following placement of orthodontic appliances.[114] The overall occurrence of white spot formation during orthodontic therapy is 11% to 12%, with little difference between bonded brackets and zinc phosphate–cemented bands.[115] A daily rinse of 0.2% NaF retards but does not completely inhibit lesion formation.[115] Unfortunately, patients most in need of significant preventive intervention are often the least compliant. Fluoride-releasing resin composites and RMGI cements inhibit demineralization adjacent to orthodontic bands and brackets. Bands cemented with glass-ionomer cements have increased retention with less demineralization compared to those cemented with zinc phosphate.[116–118] Decreased enamel demineralization under orthodontic bands retained with conventional glass-ionomer cements in vivo has been reported.[119–123]

Three prospective in vivo studies describe fluoride-releasing restorative materials used in high-caries-risk patients. Wood et al[123] placed 54 pairs of Class 5 conventional glass-ionomer (Ketac Fil) and amalgam (Sybralloy, Kerr) restorations in xerostomic patients who had been treated with radiation for head and neck cancer. When topical fluoride was not used, glass-ionomer restorations had longer survival times than amalgam restorations. When a topical NaF solution (pH 5.8) was used, amalgam restorations had a longer survival time than the glass-ionomer restorations, but recurrent caries was reduced in both groups. This study demonstrated that a fluoride-releasing material, when used without a topical fluoride gel, was effective in inhibiting recurrent caries and that acidic fluoride solutions degrade glass-ionomer restorations. It also clearly demonstrates the effectiveness of fluoride-releasing materials in noncompliant patients.

Haveman et al[124] used a similar population to measure the effectiveness of fluoride-releasing materials in high-caries-risk patients. In this study, Class 5 restorations were placed using a conventional glass ionomer (Ketac Fil), an RMGI (Vitremer), and an amalgam (Tytin, Kerr). At the 2-year recall, recurrent caries lesions had occurred at the margins of 15% of the glass-ionomer restorations, 12% of the RMGI restorations, and 44% of the amalgam restorations. Restorations with fluoride-releasing materials had significantly fewer recurrent caries lesions near the

restoration than did amalgam restorations.

McComb et al[125] reported that fluoride-releasing materials reduced the incidence of recurrent caries lesions in 45 patients who had received head and neck radiation therapy. After removal of carious tooth structure, the investigators placed Class 5 glass-ionomer (Ketac Fil), RMGI (Vitremer), and resin composite restorations. At the end of the 2-year clinical trial, subjects were classified as daily fluoride users if they used topical fluoride in trays more than 50% of the time. Topical fluoride users had no recurrent caries lesions associated with these restorations. Nonusers had an 80% reduction in recurrent caries lesions adjacent to fluoride-releasing restorations compared with the non–fluoride-releasing materials. This evidence demonstrates that if a patient uses tray-delivered supplemental fluoride, a significant reduction in caries lesions can be expected because the concentration of fluoride in the tray materials is greater than that released from the restorative material. However, if the patient does not use supplemental fluoride, then the fluoride release from the restorations still provides some level of protection for the high-caries-risk patient.

These three studies suggest that fluoride-releasing materials are somewhat effective in preventing recurrent caries lesions. However, caries lesions developed around fluoride-releasing materials in each study, demonstrating that the remineralizing effects of fluoride released from restorations can be overwhelmed if the acid challenge is great enough and that fluoride is but one factor in preventing the development of new caries lesions in high-caries-risk patients.

The American Dental Association Center for Evidence-Based Dentistry recently featured a systematic review of caries incidence at margins of glass-ionomer and amalgam restorations. The review included 10 clinical studies that compared recurrent caries at the margins of amalgam and glass-ionomer restorations of the same cavity class in the same dentition. Restorations from permanent and primary dentitions were included in the analysis. Data were pooled and compared by meta-analysis. The study revealed that there was a 65% lower chance of developing recurrent caries around single-surface glass-ionomer restorations than corresponding amalgam restorations in permanent teeth. The study found no difference in recurrent caries rates around primary teeth.[126]

However, not all studies on recurrent caries inhibition demonstrate a clear benefit from fluoride release. Tyas[127] reported no significant difference in recurrent caries in teeth restored with resin composite (Silux, 3M ESPE) or glass ionomer (Fuji II) in a population at low risk for caries. In two other clinical reports[128,129] involving almost 7,000 restorations, recurrent caries lesions developed within 10 years for

amalgam restorations, 8 years for resin composite restorations, and 5 years for glass-ionomer restorations. Recurrent caries lesions led to the replacement of almost half of the glass-ionomer restorations. In these populations, fluoride released from glass-ionomer restorations did not provide protection against primary or recurrent caries. Mjör et al[130] reported that in 9,805 replacement restorations placed by 243 Norwegian dentists in general practice, glass ionomers provided little protection against recurrent caries. Unfortunately, these retrospective studies may show that clinicians used glass ionomers in patients at high risk for caries and other materials in patients at lower risk for caries. The results in both clinical studies may be due in part to the lack of randomization of patients into the study groups and a lack of standardization in defining recurrent caries. Fluoride-releasing materials do not eliminate recurrent caries but should be viewed as one part of a complete program to reduce caries incidence.

How Much Fluoride Is Enough?

A common question is how much fluoride release from restorative materials is enough to inhibit recurrent or secondary caries. A minimum inhibitory concentration of 100 to 200 μg/mL of NaF is required to inhibit the growth of oral streptococci,[88] while 30 times that concentration is required to be bactericidal. Naturally occurring fluoride at concentrations as high as 21 μg/ mL does not produce any obvious effect on the composition of supragingival plaque. Although the fluoride levels in bacteria associated with glass-ionomer restorations are elevated, the effect of fluoride on reducing bacterial acid production, metabolism, and division has yet to be clarified.

However, another important aspect to be considered is that high levels of fluoride release produce remineralization of enamel and dentin. One study[131] demonstrated that enamel demineralization decreased as fluoride release from a resin composite restorative material increased. By extrapolating data, the authors concluded that a resin composite releasing 200 to 300 $\mu g/cm^2$ of fluoride over a 1-month period would completely inhibit secondary caries. Unfortunately, this is approximately 40 to 50 times more fluoride than is released by current fluoride-releasing resin composites. Eichmiller and Marjenhoff[64] authored an excellent review of fluoride-releasing materials and noted that caries inhibition and tooth remineralization potential have been shown in vitro by all fluoride-releasing materials when release rates were approximately 1 μg/mL. It has also been reported that fluoride concentrations of 0.03 to 0.08 ppm can shift the equilibrium from net

demineralization to net remineralization.[73]

The remineralization of dentin is more complex than that of enamel. Active dentinal caries destroys collagen matrix as well as demineralizing apatite crystals. Remineralization of dentin may be affected by the remaining mineral, the remaining collagen, or the ultrastructure of the dentin. Collagen matrix devoid of mineral does not support remineralization.[132] The difficulty in protecting root surfaces with fluoride-releasing materials may be due to the higher concentrations of fluoride needed to remineralize root surfaces compared with enamel surfaces. Wefel[132] reported that demineralized dentin, with its exposed organic matrix, did not act as a suitable matrix for remineralization and that remineralization did occur on any remaining apatite crystals. It seems clear that increased fluoride combined with calcium and phosphate is required to remineralize dentin and that the degree of remineralization that can occur in dentin may be controlled by the amount of remaining mineral content.

Based on the previous studies, it is recommended that materials with a long-term fluoride release rate of at least 2 to 3 μg/mL/day be used. With present materials, this rate of release can only be maintained when supplemental fluorides are used to recharge fluoride-releasing restorations. As has been previously stated, fluoride is only one factor needed to lower caries development, and saliva quantity and quality may be the single greatest factor in caries risk. Important proteins in saliva that stabilize calcium and phosphate are statherin, acid proline-rich proteins, and histatins. Saliva has protective properties, and when proper levels of phosphoproteins are present, calcium saturation is increased.[133]

Clinical Considerations

Although the inhibition of recurrent caries is evident for fluoride-releasing materials, their clinical effectiveness has been questioned based on the durability of the materials. Even in primary teeth, these materials should be used selectively, and the time that the material will be expected to survive based on how long the tooth will remain in the oral cavity should be evaluated against its strength and wear resistance. One report[134] describes the wear resistance of a compomer (Dyract) and shows that this material has poorer wear resistance than resin composite. At 1 year in a clinical evaluation of 91 Dyract restorations in conservative Class 1 and Class 2 restorations in primary teeth, the mean wear of Dyract was 190 μm, compared with approximately 10 μm for a wear-resistant resin composite. Another study[135]

measuring marginal adaptation of Class 2 restorations in the permanent dentition reported that compomers showed poor adaptation after 6 months of clinical wear, prompting the authors to recommend that compomers not be used as definitive restorations in load-bearing areas. Although the absolute numbers may vary, the typical compomer undergoes greater wear than the typical resin composite. A 4-year clinical evaluation of posterior compomer restorations revealed a 16% failure rate.[136] Therefore, compomers should not be used in Class 1 or Class 2 load-bearing areas in the permanent dentition. However, as the wear resistance and fluoride recharge of compomer restorative materials continue to improve, it is now apparent that they can be used with success in both Class 5 and Class 2 open sandwich restorations. Three in vivo studies[137–139] have reported the clinical success of Dyract as a Class 5 restorative material in noncarious cervical lesions. In these studies, the compomer was clinically acceptable at the end of 3 years[137] and was superior to RMGIs.[138,139] Though compomers have improved, much clinical testing is necessary to document their clinical performance before they can be endorsed for more expanded applications.

Treating the high-caries-risk patient requires special selection of restorative materials. RMGIs and compomers are recommended as the esthetic restorative materials of choice in Class 5 restorations in xerostomic patients and other patients at high risk for caries. This is because of the fluoride-release and fluoride-recharge capabilities of these materials. Because RMGIs and compomers have poorer wear resistance than resin composites, a resin composite should be used for restorations in load-bearing areas.

Another strategy for Class 2 and Class 5 restorations that extend gingivally below the cementoenamel junction is the open sandwich technique (Fig 14-9). In this technique, the fluoride-releasing material is placed along the gingival margin of a preparation (the proximal box of a Class 2 preparation) when the gingival margin extends to cementum or dentin. The RMGI is cured, and resin composite is placed in increments to restore the occlusal surface for adequate wear resistance (Fig 14-10). The open sandwich technique was clinically successful at the 5-year recall[140] of Class 2 composite restorations placed in patients with moderate to high caries risk. An RMGI (Vitremer) was placed in the gingival portion of the proximal box and covered with a resin composite (Z100, 3M ESPE). Another 6-year study found acceptable durability of RMGIs in open sandwich restorations.[141] The open sandwich technique in Class 2 restorations combines the wear resistance of a resin composite with the fluoride release and recharge potential of RMGIs. Conventional glass ionomers are less successful in the open sandwich technique.[142] Because of

increased fluoride release compared with that of resin composite, better handling, and a likely enhanced solubility resistance in an acidic environment, compomers have been suggested for this application. However, a 9-year clinical trial found no difference in the incidence of secondary caries using a compomer in an open sandwich technique versus a resin composite alone.[143]

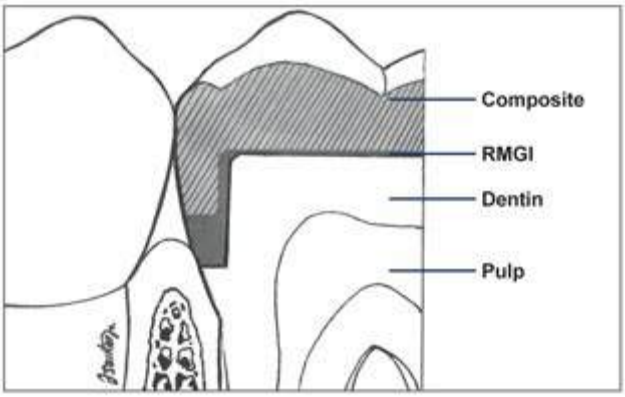


Fig 14-9 Illustration of the open sandwich technique.

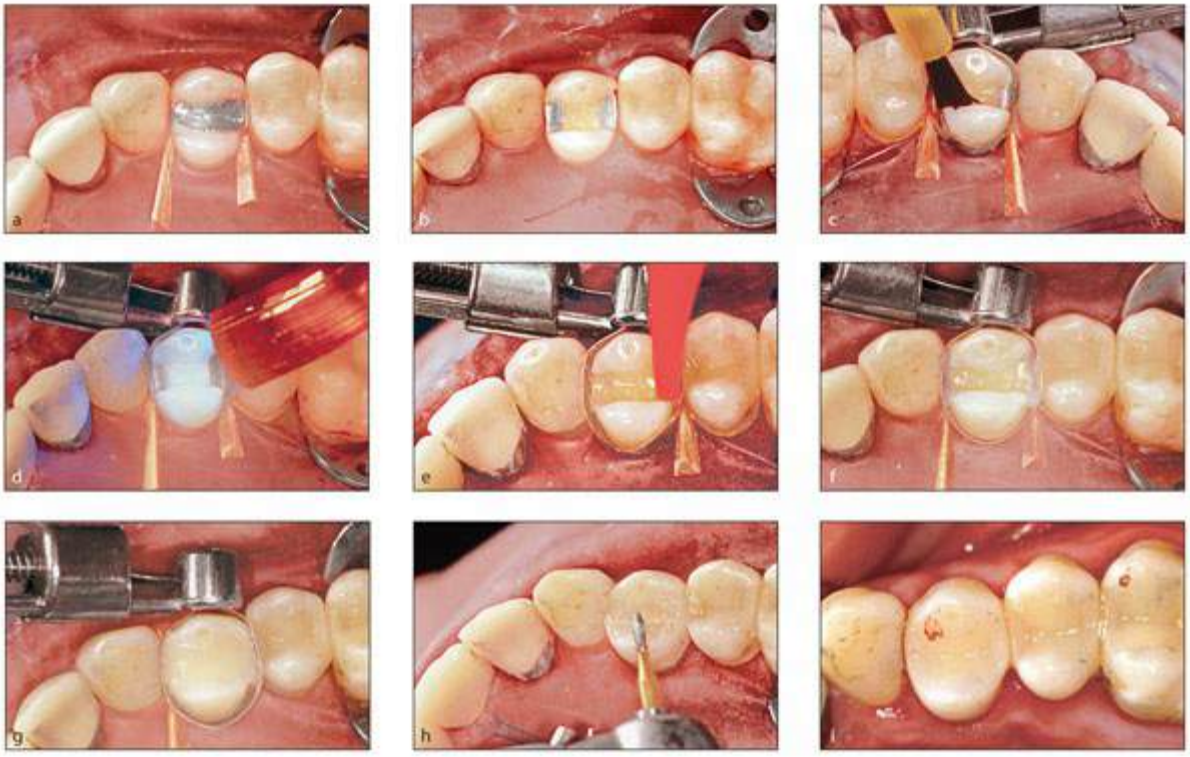


Fig 14-10 Open sandwich technique. The fluoride-releasing RMGI material is placed in the proximal box. A wear-resistant composite is placed on the occlusal surface. *(a)* Defective amalgam. *(b)* Cavity preparation. *(c and d)* Primer application and light curing of the primer. *(e)* RMGI placement. *(f)* Etching in preparation for adhesive and composite placement. Note that the RMGI is placed as a thin layer, apical to proximal contacts only. *(g)* Resin composite placement. *(h)* Finishing. *(i)* Definitive restoration.

As dental restorative materials continue to proliferate, it becomes increasingly difficult to choose an appropriate material for a particular clinical situation. Fluoride-releasing materials are no exception, and clinicians need guidelines to select and use these materials effectively (Table 14-3). Today, there is modest but growing evidence from clinical trials that fluoride-releasing materials, especially RMGIs and conventional glass ionomers, reduce recurrent caries. There is also evidence of a dose-response relationship between fluoride release and decreased caries. While materials with higher fluoride-releasing capability have a greater caries-inhibiting effect, these materials are not panaceas. The physical and mechanical limitations of glass ionomers and compomers, especially poor wear resistance[144–146]; staining and discoloration with products such as coffee, tea, wine, or exposure to daylight[147]; and maintenance of surface gloss,[147] contribute markedly to restoration failure (Figs 14-11 to 14-13).

Table 14-3	**Guidelines for recommended use of fluoride-releasing materials**
Glass ionomer	Provisional restorations or caries control for patients with high caries risk; Class 3 and Class 5 restorations; cores or buildups when half or more of the tooth remains; ART; sealants for erupting teeth
RMGI	Provisional restorations or caries control for patients with high caries risk; patients with moderate salivary flow; Class 3 and Class 5 restorations; buildups when half or more of the tooth remains; ART; open sandwich technique; sealants for erupting teeth
Giomer/compomer	High-caries-risk patients with diminished salivary flow; primary teeth; permanent Class 3 and Class 5 restorations.
Fluoride-releasing composite	Long-term provisional restorations; conservative Class 1, 2, 3, 4, 5 restorations; core buildups

ART—atraumatic restorative technique.

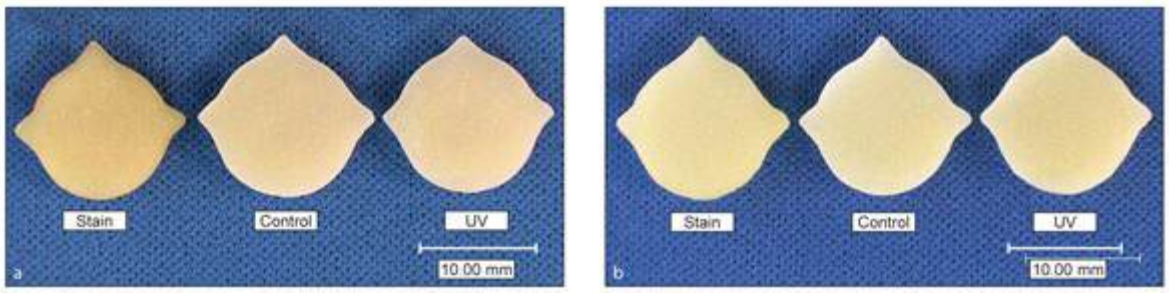


Fig 14-11 Marked color differences are seen with RMGI products (Fuji II LC A2) *(a)* compared with resin composites (Clearfil Majesty A2, Kuraray Dental) following immersion in a solution of

cranberry juice, coffee, and tea *(b)*.

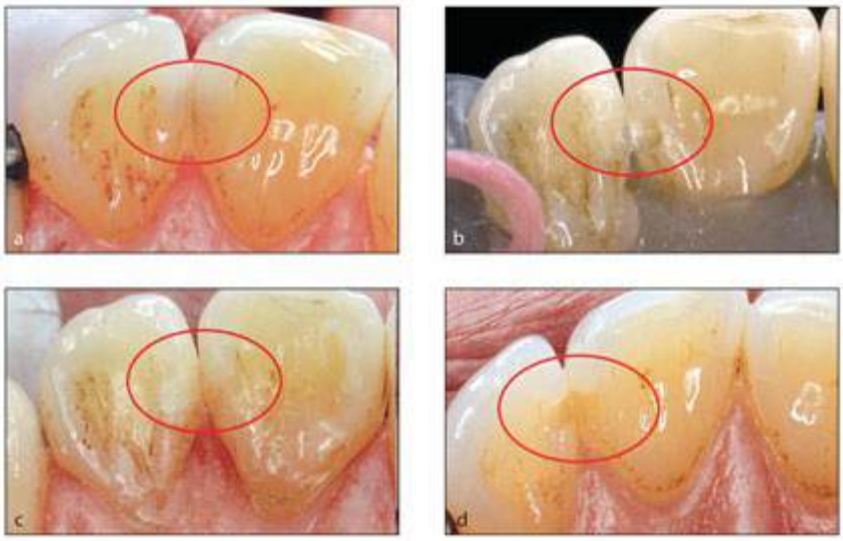


Fig 14-12 Clinical representation of a stained glass-ionomer restoration. *(a)* Caries. *(b)* Class 3 cavity preparation. *(c)* Restoration at baseline. *(d)* Restoration at 1-year recall. Note the shade difference compared with *(c)*.



Fig 14-13 Class 5 glass-ionomer restorations. The left central and right lateral incisors contain 2-year-old restorations, whereas the right central incisor contains a newly placed restoration of the same shade. Note the evident staining of the older restorations.

Fluoride-releasing materials should be used in high-caries-risk patients as one component of their overall treatment. High-caries-risk patients also require oral hygiene counseling, diet modification, saliva-replacement materials, and daily application of neutral NaF-containing toothpastes with added calcium and phosphate and xylitol to inhibit caries activity. An absolute requirement for these patients is frequent recall and site-specific quarterly applications of fluoride varnishes.

Generally a 5% NaF-containing varnish should be used in hard-to-clean areas that are prone to white spot lesion development. For patients with low salivary output and white spot lesion development, a regimen of calcium phosphate–containing material, such as MI Paste Plus (GC America) (a casein phospho-peptide with amorphous calcium phosphate–containing paste with 900 ppm fluoride), is also recommended to increase levels of calcium and phosphate in the saliva and promote remineralization. These recommendations, combined with frequent oral hygiene checks and reinforcement, should provide maximum benefit for the individual who is at high risk for caries.[148–149] See chapter 5 for more on managing high-caries-risk patients.

References

1. Niessen LC, Gibson G. Oral health for a lifetime: Preventive strategies for the older adult. Quintessence Int 1997;28:626–630.
2. Newbrun E. Topical fluorides in caries prevention and management: A North American perspective. J Dent Educ 2001;65:1078–1083.
3. Tenuta LM, Cury JA. Fluoride: Its role in dentistry. Braz Oral Res 2010;24(suppl 1):9–17.
4. Loesche WJ, Hockett, RN, Syed SA. The predominant cultivable flora of tooth surface plaque removed from institutionalized subjects. Arch Oral Biol 1972;17:1311–1325.
5. Douglass JM, Li Y, Tinanoff N. Association of mutans streptococci between caregivers and their children. Pediatr Dent 2008;30:375–387.
6. Featherstone JD. The caries balance: Contributing factors and early detection. J Calif Dent Assoc 2003;31:129–133.
7. Van Dorp CS, Ten Cate JM. Preventive measures and caries progression: An in vitro study on fissures and smooth surfaces of human molars. ASDC J Dent Child 1992;59:257–262.
8. ten Cate JM, Buijs MJ, Damen JJ. pH-cycling of enamel and dentin lesions in the presence of low concentrations of fluoride. Eur J Oral Sci 1995;103:362–367.
9. McIntyre J. The nature and progression of dental caries. In: Mount EJ, Hume WR (eds). Preservation and Restoration of Tooth Structure. London: Mosby, 1998:20–23.
10. Seppä L, Korhonen A, Nuutinen A. Inhibitory effect on S. mutans by fluoride-treated conventional and resin-reinforced glass ionomer cements. Eur J Oral Sci 1995;103:182–185.
11. Bardow A, Nyvad B, Nauntofte B. Relationships between medication intake, complaints of dry mouth, salivary flow rate and composition, and the rate of tooth demineralization in situ. Arch Oral Biol 2001;46: 413–423.
12. Billings RJ, Proskin HM, Moss ME. Xerostomia and associated factors in a community-dwelling adult population. Community Dent Oral Epidemiol 1996;24:312–316.
13. Meurman JH, ten Cate JM. Pathogenesis and modifying factors of dental erosion. Eur J Oral Sci 1996;104:199–206.
14. Norman RD, Platt JR, Phillips RW, Swartz ML. Additional studies on fluoride uptake by enamel from certain dental materials. J Dent Res 1961;40:529–537.
15. Yip HK, Smales RJ, Ngo HC, Tay FR, Chu FC. Selection of restorative materials for the atraumatic restorative treatment (ART) approach: A review. Spec Care Dentist 2001;21:216–221.
16. Lawson NC, Cakir D, Ramp LC, Beck P, Burgess J. Effect of light activation on resin-modified glass ionomer shear bond strength. Oper Dent 2012; 37:380–385.

17. Mousavinasab SM, Meyers J. Fluoride release by glass ionomer cements, compomer and giomer. Dent Res J (Isfahan) 2009;6:75–81.
18. Dhull KS, Nandlal B. Comparative evaluation of fluoride release from PRG-composites and compomer on application of topical fluoride: An in-vitro study. J Indian Soc Pedod Prev Dent 2009;27:27–32.
19. Manuja N, Pandit IK, Srivastava N, Gugnani N, Nagpal R. Comparative evaluation of shear bond strength of various esthetic restorative materials to dentin: An in vitro study. J Indian Soc Pedod Prev Dent 2011;29: 7–13.
20. McKenzie MA, Linden RW, Nicholson JW. The physical properties of conventional and resin-modified glass-ionomer dental cements stored in saliva, proprietary acidic beverages, saline and water. Biomater 2003; 24:4063–4069.
21. Ilie N, Hickel R. Mechanical behavior of glass ionomer cements as a function of loading condition and mixing procedure. Dent Mater 2007;26: 526–533.
22. Lien W, Vandewalle KS. Physical properties of a new silorane-based restorative system. Dent Mater 2010;26:337–344.
23. Almuammar MF, Schulman A, Salama FS. Shear bond strength of six restorative materials. J Clin Pediatr Dent 2001;25:221–225.
24. Lawson NC, Burgess JO. Evaluation of the bond strength with the immediate dentinal sealing technique [abstract 3126]. J Dent Res 2008;87 (special issue B).
25. Berzins DW, Abey S, Costache MC, Wilkie CA, Roberts HW. Resin-modified glass-ionomer setting reaction competition. J Dent Res 2010;89:82–86.
26. Coutinho E, Cardoso MV, De Munck J, et al. Bonding effectiveness and interfacial characterization of a nanofilled resin-modified glass-ionomer. Dent Mater 2009;25:1347–1357.
27. Burgess J, Cakir D. Comparative properties of low-shrinkage composite resins. Compend Contin Educ Dent 2010;31(special issue):10–15.
28. Gerdolle DA, Mortier E, Droz D. Microleakage and polymerization shrinkage of various polymer restorative materials. J Dent Child 2008; 75:125–133.
29. Attin T, Buchalla W, Kielbassa AM, Helwig E. Curing shrinkage and volumetric changes of resin-modified glass ionomer restorative materials. Dent Mater 1995;11:359–362.
30. Dauvillier BS, Feilzer AJ, De Gee AJ, Davidson CL. Visco-elastic parameters of dental restorative materials during setting. J Dent Res 2000;79: 818–823.
31. Feilzer AJ, Kakaboura AI, De Gee AJ, Davidson CL. The influence of water sorption on the development of setting shrinkage stress in traditional and resin-modified glass ionomer cements. Dent Mater 1995;11: 186–190.
32. Park JB, Lakes RS. Biomaterials, ed 2. New York: Plenum, 1992.
33. Cardenas HL, Burgess JO. Thermal expansion of glass ionomers [abstract 946]. J Dent Res 1994;73(special issue):220.
34. Koch G, Hatibovic-Kofman S. Glass ionomer cements as a fluoride release system in vivo. Swed Dent J 1990;14:267–273.
35. Hadley PC, Milella E, Gerardi C, Hill RG, Billington RW. Distribution of fluoride in glass ionomer cement determined using SIMS. Biomater 2001;22:1563–1569.
36. Arends J, Dijkman GE, Dijkman AG. Review of fluoride release and secondary caries reduction by fluoridating composites. Adv Dent Res 1995;9:367–76.
37. Ruse ND. What is a “compomer”? J Can Dent Assoc 1999;65:500–504.
38. Norling B, Summitt J. Resin ionomer restorative materials: The new generation. J Esthet Dent 1994;6:207–215.
39. Walls AW, Mather E. Fuji IX restorations in occlusal cavities of molar teeth, 12-month data [abstract 1443]. J Dent Res 1999;78:286.
40. Robles R, Kalavacharla VK, Beek P, et al. Wear of glass ionomer restorative materials [abstract 865]. Presented at the 42nd Annual Meeting of the American Association of Dental Research, Tampa, 23 March 2012.
41. Mazer RB, Leinfelder KF. Evaluating a microfill posterior composite resin. A five-year study. J Am Dent Assoc 1992;123:33–38.

42. Barkmeier WW, Latta MA, Erickson RL, Wilwerding TM. Wear simulation of resin composites and the relationship to clinical wear. Oper Dent 2008;33:177–182.
43. Frankenberger R, Garcia-Godoy F, Krämer N. Clinical performance of viscous glass ionomer cement in posterior cavities over two years. Int J Dent 2009:781462.
44. Coutinho E, Yoshida Y, Inoue S, et al. Gel phase formation at resin-modified glass-ionomer/tooth interfaces. J Dent Res 2007;86:656–661.
45. Xu X, Burgess JO. Compressive strength, fluoride release and recharge of fluoride-releasing materials. Biomater 2003;24:2451–2461.
46. Johnson GH, Lepe X, Zhang H, Wataha JC. Retention of metal-ceramic crowns with contemporary dental cements. J Am Dent Assoc 2009; 140:1125–1136.
47. Daou MH, Tavernier B, Meyer JM. Clinical evaluation of four different dental restorative materials: One-year results. Schweiz Monatsschr Zahnmed 2008;118:290–295.
48. Elderton RJ, Aboush Y, Vowles RW, Bell CJ, Marshall KJ. Retention of cervical Dyract compomer restorations after two years [abstract 49]. J Dent Res 1996;75:24.
49. Gallo JR, Burgess JO, Ripps AH, et al. Three-year clinical evaluation of a compomer and a resin composite as class V filling materials. Oper Dent 2005;30:275–281.
50. Hermesch CB, Wall BS, McEntire JF. Dimensional stability of dental restorative materials and cements over four years. Gen Dent 2003;51: 518–523.
51. Milward PJ, Adusei GO, Lynch CD. Improving some selected properties of dental polyacid-modified composite resins. Dent Mater 2011;27: 997–1002.
52. Ikemura K, Tay FR, Endo T, Pashley DH. A review of chemical-approach and ultramorphological studies on the development of fluoride-releasing dental adhesives comprising new pre-reacted glass ionomer (PRG) fillers. Dent Mater 2008;27:315–339.
53. Gordan VV, Mondragon E, Watson RE, Garvan C, Mjör IA. A clinical evaluation of a self-etching primer and a giomer restorative material: Results at eight years. J Am Dent Assoc 2007;138:621–627.
54. Naoum S, Ellakwa A, Martin F, Swain M. Fluoride release, recharge and mechanical property stability of various fluoride-containing resin composites. Oper Dent 2011;36:422–432.
55. Hals E, Norderval IT. Histopathology of experimental in vivo caries around silicate fillings. Acta Odontol Scand 1973;31:357–367.
56. Phillips RW, Swartz ML. Effect of certain restorative materials on solubility of enamel. J Am Dent Assoc 1957;54:623–636.
57. Norman RD, Phillips RW, Swartz ML. Fluoride uptake by enamel from certain dental materials. J Dent Res 1960;39:11–16.
58. Robertson MA, Kau CH, English JD, Lee RP, Powers J, Nguyen JT. MI Paste Plus to prevent demineralization in orthodontic patients: A prospective randomized controlled trial. Am J Orthod Dentofacial Orthop 2011;140:660–668.
59. Norman RD, Mehra RV, Swartz ML, Phillips RW. Effects of restorative materials on plaque composition. J Dent Res 1972;51:1596–1601.
60. Perrin C, Persin M, Sarrazin J. A comparison of fluoride release from four glass-ionomer cements. Quintessence Int 1994;25:603–608.
61. Alvarez AN, Burgess JO, Chan DCN. Short-term fluoride release of six ionomers—Recharged, coated and abraded [abstract 260]. J Dent Res 1994;73:134.
62. Burkett L, Burgess JO, Chan DCN, Norling BK. Fluoride release of glass ionomers coated and not coated with adhesive [abstract 1242]. J Dent Res 1993;72:258.
63. Cranfield M, Kuhn AT, Winter GB. Factors relating to the rate of fluorideion release from glass-ionomer cement. J Dent 1982;10:333–341.
64. Eichmiller FC, Marjenhoff WA. Fluoride-releasing dental restorative materials. Oper Dent 1998;23:218–228.
65. Forsten L, Paunio IK. Fluoride release by silicate cements and composite resins. Scand J Dent Res 1972;80:515–519.
66. Forsten L. Fluoride release of glass ionomers. J Esthet Dent 1994;6: 216–222.

67. Forsten L. Short- and long-term fluoride release from glass ionomers and other fluoride-containing filling materials in vitro. Scand J Dent Res 1990;98:179–185.
68. Forsten L. Fluoride release from a glass ionomer cement. Scand J Dent Res 1977;85:503–504.
69. McKnight-Hanes C, Whitford GM. Fluoride release from three glass ionomer materials and the effects of varnishing with or without finishing. Caries Res 1992;26:345–350.
70. Mitra SB. In vitro fluoride release from a light-cured glass-ionomer liner/ base. J Dent Res 1991;70:75–78.
71. Hsu HM, Huang GF, Chang HH, Wang YL, Guo MK. A continuous flow system for assessing fluoride release/uptake of fluoride-containing restorative materials. Dent Mater 2004;20:740–749.
72. Woolford MJ, Grieve AR. Release of fluoride from glass polyalkenoate (ionomer) cement subjected to radiant heat. J Dent 1995;23:233–237.
73. Wiegand A, Buchalla W, Attin T. Review on fluoride-releasing restorative materials—Fluoride release and uptake characteristics, antibacterial activity and influence on caries formation. Dent Mater 2007;23:343–362.
74. De Moor RJ, Verbeeck RM, De Maeyer EA. Fluoride release profiles of restorative glass ionomer formulations. Dent Mater 1996;12:88–95.
75. Hatibovic-Kofman S, Koch G. Fluoride release from glass ionomer cement in vivo and in vitro. Swed Dent J 1991;15:253–258.
76. Hattab FN, el-Mowafy OM, Salem NS, el-Badrawy WA. An in vivo study on the release of fluoride from glass-ionomer cement. Quintessence Int 1991;22:221–224.
77. Moreau JL, Xu HH. Fluoride releasing restorative materials: Effects of pH on mechanical properties and ion release. Dent Mater 2010;26: e227–e235.
78. Hallgren A, Oliveby A, Twetman S. Fluoride concentration in plaque adjacent to orthodontic appliances retained with glass ionomer cement. Caries Res 1993;27:51–54.
79. Forsten L, Rytömaa I, Anttilla A, Keinonen J. Fluoride uptake from restorative dental materials by human enamel. Scand J Dent Res 1976;84: 391–395.
80. Vercruysse CW, De Maeyer EA, Verbeeck RM. Fluoride release of polyacid-modified composite resins with and without bonding agents. Dent Mater 2001;17:354–358.
81. Tam LE, Chan GP, Yim D. In vitro caries inhibition effects by conventional and resin-modified glass-ionomer restorations. Oper Dent 1997; 22:4–14.
82. Hotta M, Hirukawa H. Abrasion resistance of restorative glass-ionomer cements with a light-cured surface coating. Oper Dent 1994;19:42–46.
83. Kupietzky A, Houpt M, Mellberg J, Shey Z. Fluoride exchange from glass ionomer preventive resin restorations. Pediatr Dent 1994;16:340–345.
84. Forsten L. Fluoride release and uptake by glass ionomers. Scand J Dent Res 1991;99:241–245.
85. Tantbirojn D, Douglas WH, Versluis A. Inhibitive effect of a resin-modified glass ionomer cement on remote enamel artificial caries. Caries Res 1997;31:275–280.
86. Damen JJ, Buijs MJ, ten Cate JM. Uptake and release of fluoride by saliva-coated glass ionomer cement. Caries Res 1996;30:454–457.
87. Diaz-Arnold AM, Holmes DC, Wistrom DW, Swift EJ. Short-term fluoride release/uptake of glass ionomer restoratives. Dent Mater 1995;11:96–101.
88. Ekstrand J, Fejerskov O, Silverstone LM (eds). Fluoride in Dentistry. Copenhagen: Munksgaard, 1988:76.
89. Preston AJ, Higham SM, Agalamanyi EA, Mair LH. Fluoride recharge of aesthetic dental materials. J Oral Rehabil 1999;26:936–940.
90. Burgess JO, Cardenas HL. Surface roughness of topical fluoride treated glass ionomers [abstract 775]. J Dent Res 1995;74:108.
91. Diaz-Arnold AM, Wistrom DW, Swift EJ. Topical fluoride and glass ionomer microhardness. Am J Dent 1995;8:134–136.
92. El-Badrawy WA, McComb D. Effect of home-use fluoride gels on resin-modified glass-ionomer cements. Oper Dent 1998;23:2–9.
93. Papagiannoulis L, Tzoutzas J, Eliades G. Effect of topical fluoride agents on the morphologic characteristics and composition of resin composite restorative materials. J Prosthet Dent 1997;77:405–413.

94. Prati C, Tao L, Simpson M, Pashley DH. Permeability and microleakage of class II resin composite restorations. J Dent 1994;22:49–56.
95. Mjör IA, Toffenetti F. Secondary caries: A literature review with case reports. Quintessence Int 2000;31:165–179.
96. Özer L. The Relationship Between Gap Size, Microbial Accumulation and the Structural Features of Natural Caries in Extracted Teeth With Class II Amalgam Restorations [thesis]. Copenhagen: University of Copenhagen, 1997.
97. Kidd EA, Beighton D. Prediction of secondary caries around tooth-colored restorations: A clinical and microbiological study. J Dent Res 1996;75:1942–1946.
98. Kidd EA, Fejerskov O. What constitutes dental caries? Histopathology of carious enamel and dentin related to the action of cariogenic bio-film. J Dent Res 2004;83(suppl 1):C35–C38.
99. Benelli EM, Serra MC, Rodrigues AL Jr, Cury JA. In situ anticariogenic potential of glass ionomer cement. Caries Res 1993;27:280–284.
100. Loyola-Rodriguez JP, Garcia-Godoy F, Lindquist R. Growth inhibition of glass ionomer cements on mutans streptococci. Pediatr Dent 1994; 16:346–349.
101. Forss H, Jokinen J, Spets-Happonen S, Seppä L, Luoma H. Fluoride and mutans streptococci in plaque grown on glass ionomer and composite. Caries Res 1991;25:454–458.
102. Forss H, Nase L, Seppä L. Fluoride and caries associated microflora in plaque on old GIC fillings [abstract 2468]. J Dent Res 1994;73:410.
103. Palenik CJ, Behnen MJ, Setcos JC, Miller CH. Inhibition of microbial adherence and growth by various glass ionomers in vitro. Dent Mater 1992;8:16–20.
104. McComb D, Erickson D. Antimicrobial action of new, proprietary lining cements. J Dent Res 1987;66:1025–1028.
105. DeSchepper EJ, White RR, von der Lehr W. Antibacterial effects of glass ionomers. Am J Dent 1989;2:51–56.
106. DeSchepper EJ, Thrasher MR, Thurmond BA. Antibacterial effects of light-cured liners. Am J Dent 1989;2:74–76.
107. Al-Naimi OT, Itota T, Hobson RS, McCabe JF. Fluoride release for restorative materials and its effect on biofilm formation in natural saliva. J Mater Sci Mater Med 2008;19:1243–1248.
108. Svanberg M, Mjör IA, Orstavik D. Mutans streptococci in plaque from margins of amalgam, composite, and glass-ionomer restorations. J Dent Res 1990;69:861–864.
109. Van Dijken JW, Kalfas S, Litra V, Oliveby A. Fluoride and mutans streptococci levels in plaque on aged restorations of resin-modified glass ionomer cement, compomer and resin composite. Caries Res 1997; 31:379–383.
110. Frencken JE, Makoni F, Sithole WD. ART restorations and glass ionomer sealants in Zimbabwe: Survival after 3 years. Community Dent Oral Epidemiol 1998;26:372–381.
111. O'Reilly MM, Featherstone JD. Demineralization and remineralization around orthodontic appliances: An in vivo study. Am J Orthod Dento-facial Orthop 1987;92:33–40.
112. Øgaard B, Rølla G, Arends J, ten Cate JM. Orthodontic appliances and enamel demineralization. Part 2. Prevention and treatment of lesions. Am J Orthod Dentofacial Orthop 1988;94:123–128.
113. Rezk-Lega F, Øgaard B, Rølla G. Availability of fluoride from glass-ionomer luting cements in human saliva. Scand J Dent Res 1991;99: 60–63.
114. Lundström F, Krasse B. Streptococcus mutans and lactobacilli frequency in orthodontic patients: The effect of chlorhexidine treatments. Eur J Orthod 1987;9:109–116.
115. Gorelick L, Geiger A, Gwinnett A. Incidence of white spot formation after bonding and banding. Am J Orthod 1982;81:93–98.
116. Fricker JP, McLachlan MD. Clinical studies of glass ionomer cements. Part 2—A two-year clinical study comparing glass ionomer cement with zinc phosphate cement. Aust Orthod J 1987;10:12–14.
117. Norris DS, Mclnnes-Ledoux P, Schwaninger B, Weinberg R. Retention of orthodontic bands with new fluoride-releasing cements. Am J Orthod 1986;89:206–211.

118. Stirrups DR. A comparative clinical trial of a glass ionomer and a zinc phosphate cement for securing orthodontic bands. Br J Orthod 1991; 18:15–20.
119. Kvam E, Broch J, Nissen-Meyer JH. Comparison between a zinc phosphate cement and a glass ionomer cement for cementation of orthodontic bands. Eur J Orthod 1983;5:307–313.
120. Marcusson A, Norevall LI, Persson M. White spot reduction when using glass ionomer cement for bonding in orthodontics: A longitudinal and comparative study. Eur J Orthod 1997;19:233–242.
121. Ortendahl T, Thilander B, Svanberg M. Mutans streptococci and incipient caries adjacent to glass ionomer cement or resin-based composite in orthodontics. Am J Orthod Dentofacial Orthop 1997;112:271–274.
122. Seeholzer HW, Dasch W. Banding with a glass ionomer cement. J Clin Orthod 1988;22:165–169.
123. Wood RE, Maxymiw WG, McComb D. A clinical comparison of glass ionomer (polyalkenoate) and silver amalgam restorations in the treatment of class 5 caries in xerostomic head and neck cancer patients. Oper Dent 1993;18:94–102.
124. Haveman C, Burgess J, Summitt JB. A clinical comparison of restorative materials for caries in xerostomic patients [abstract 1441]. J Dent Res 1999;78:286.
125. McComb D, Erickson RL, Maxymiw WG, Wood RE. A clinical comparison of glass-ionomer, resin-modified glass ionomer and resin-composite restorations in the treatment of cervical caries in xerostomic head and neck radiation patients. Oper Dent 2002;27:430–437.
126. Mickenautsch S, Yengopal V. Absence of carious lesions at margins of glass-ionomer cement and amalgam restorations: An update of systematic review evidence. BMC Res Notes 2011;4:58.
127. Tyas MJ. Cariostatic effect of glass ionomer cement: A five-year clinical study. Aust Dent J 1991;36:236–239.
128. Mjör IA, Jokstad A. Five-year study of class II restorations in permanent teeth using amalgam, glass polyalkenoate (ionomer) cermet and resin-based composite materials. J Dent 1993;21:338–343.
129. Mjör IA. Glass-ionomer cement restorations and secondary caries: A preliminary report. Quintessence Int 1996;27:171–174.
130. Mjör IA, Moorhead JE, Dahl JE. Reasons for replacement of restorations in permanent teeth in general dental practice. Int Dent J 2000;50: 361–366.
131. Dijkman GE, de Vries J, Lodding A, Arends J. Long-term fluoride release of visible light-activated composites in vitro: A correlation with in situ demineralisation data. Caries Res 1993;27:117–123.
132. Wefel JS. Root caries histopathology and chemistry. Am J Dent 1994; 7:261–265.
133. Walsh LJ. Evidence that demands a verdict: Latest developments in remineralization therapies. Australasian Dent Pract 2009;20:48–59.
134. Peters TC, Roeters JJ, Frankenmolen FW. Clinical evaluation of Dyract in primary molars: 1-year results. Am J Dent 1996;9:83–88.
135. Kersten S, Besek MJ, Lutz F. Marginal adaptation of class-II-compomer restorations in vivo [abstract 2286]. J Dent Res 1999;78:391.
136. Krämer N, Garcia-Godoy F, Reinelt C, Frankenberger R. Clinical performance of posterior compomer restorations over 4 years. Am J Dent 2006;19:61–66.
137. Martin N, Jedynakiewicz NM, Fletcher JM. A clinical evaluation of a new compomer restorative at 3 years in the restoration of non-retentive cervical cavities [abstract]. J Dent Res 2002;82:A52.
138. Abdalla Al, Alhadainy HA, Garcia-Godoy F. Clinical evaluation of glass ionomers and compomers in class V carious lesions. Am J Dent 1997; 10:18–20.
139. Loher C, Kunzelmann RH, Hickel R. Clinical evaluation of glass ionomer cements (LC), compomer and composite restorations in class V cavities—Two year results [abstract 1190]. J Dent Res 1997;76:162.
140. Burgess JO, Summitt JB, Robbins JW, et al. Clinical evaluation of base, sandwich and bonded Class 2 resin composite restorations [abstract 304]. J Dent Res 1999;78:531.
141. Andersson-Wenckert IE, van Dijken JW, Kieri C. Durability of extensive Class II open-sandwich restorations with a resin-modified glass ionomer cement after 6 years. Am J Dent 2004;17:43–50.
142. Welbury RR, Murray JJ. A clinical trial of the glass-ionomer cement-composite resin “sandwich” technique in Class II cavities in permanent premolar and molar teeth. Quintessence Int 1990;21:507–512.

143. Lindberg A, van Dijken JW, Lindberg M. Nine-year evaluation of a polyacid-modified resin composite/resin composite open sandwich technique in Class II cavities. J Dent 2007;35:124–129.
144. Da Cunha MR, Puppin-Rontani RM, Ferracane JL, Correr-Sobrinho L. In vitro wear evaluation of dental materials in primary teeth. Am J Dent 2006;19:364–369.
145. Correr GM, Bruschi Alonso RC, Correr Sobrinho L, Puppin-Rontani RM, Ferracane JL. In vitro wear of resin-based materials—Simultaneous corrosive and abrasive wear. J Biomed Mater Res B Appl Biomater 2006;78:105–114.
146. Shabanian M, Richards LC. In vitro wear rates of materials under different loads and varying pH. J Prosthet Dent 2002;87:650–656.
147. Vance M, Cakir D, Beck P, Litaker MS, Burgess J. Color and gloss of direct esthetic restorative materials [abstract 1505]. J Dent Res 2009; 88(special issue A).
148. Reynolds EC. Anticariogenic complexes of amorphous calcium phosphate stabilized by casein phosphopeptides: A review. Spec Care Dentist 1998;18:8–16.
149. Shen P, Cai F, Walker G, Reynolds C, Reynolds EC. Enamel remineralization by a mouthrinse containing casein phosphopeptide-amorphous calcium phosphate and fluoride in an in situ model. Aust Dent J 2004;49(ADRF special research suppl):4.